

Midterm exam of Statistics & Probability (2^h)

Name :

Groupe:

Exercise 1. Among these assertions, specify which ones are true and which ones are false. (0.25x20)

	True	False
1-The geometric mean is always less than or equal to the arithmetic mean.	✓	
2-The range is a measure of central tendency.		✓
3-The median is not affected by the presence of extreme values in a dataset.	✓	
4-The STD of a sample can be -1.		✓
5-The sum of deviations of data points from the mean is always zero	✓	
6-The mode can be used for both numerical and categorical data	✓	
7-The mean is equal to 0 for a symmetric histogram.		✓
8-If a value is higher than the mean then it is an outlier.		✓
9-Percentiles divide data into 100 equal parts	✓	
10-The sample mean is always higher than the population mean.		✓
11- $\rho(X, Y)$ is a unitless number between -1 and 1.	✓	
12-The coefficient of skewness is zero for a perfectly symmetric distribution.	✓	
13-Parameter is a numerical measure that describes a variable of a population.	✓	
14-The harmonic mean is most appropriate for datasets with large outliers.	✓	
15-Chebyshev's theorem applies to any set of data.	✓	
16-When $\rho(X, Y)$ is close to zero, it indicates a weak linear relationship.	✓	
17-A correlation of 1 or -1 means that the two variables are perfectly linearly related.	✓	
18-A Box-plot can detect the outliers.	✓	
19-The interquartile range is the difference between the first and second quartiles.		✓
20- $\forall a; b; x_0; y_0 \in \mathbb{R} : Cov(aX + x_0; bY + y_0) \neq abCov(X; Y)$		✓

Exercise 2. A teacher has collected exam scores from a class of 40 students and grouped them into classes. The data is as follows:

(1)

Class (Score)	Mid-point (x_i)	frequency (f_i)	Rel. Freq.	Cum. Freq.	Cum. Rel. Freq.
50 – 59	$\frac{50+59}{2} = 54.5$	5	$\frac{5}{40} = 0.125$	5	0.125
60 – 69	64.5	$0.2 \times 40 = 8$	0.2	13	0.325
70 – 79	74.5	12	0.3	25	0.625
80 – 89	84.5	10	0.25	35	0.875
90 – 99	94.5	5	0.125	40	1
Σ		40			

- Complete the statistical table above.
- Calculate the geometric, Harmonic and arithmetic mean score then compare them.

Geometric mean score :

$$GM = \sqrt[5]{\prod_{i=1}^5 x_i^{f_i}} = \sqrt[40]{54.5^5 \times 64.5^8 \times 74.5^{12} \times 84.5^{10} \times 94.5^5} = 74.003 \quad (0.5)$$

Harmonic mean score :

$$HM = \frac{\sum_{i=1}^5 f_i}{\sum_{i=1}^5 \frac{f_i}{x_i}} = \frac{40}{\frac{5}{54.5} + \frac{8}{64.5} + \frac{12}{74.5} + \frac{10}{84.5} + \frac{5}{94.5}} = \mathbf{72.979} \quad (0.5)$$

Arithmetic mean score :

$$AM = \bar{X} = \frac{1}{\sum_{i=1}^5 f_i} \sum_{i=1}^5 f_i x_i = \frac{5 \times 54.5 + 8 \times 64.5 + 12 \times 74.5 + 10 \times 84.5 + 5 \times 94.5}{40} = \mathbf{75} \quad (0.5)$$

and we have

$$HM \leq GM \leq AM \quad (0.25)$$

3. Identify the modal class and determine the Mode

The highest frequency is for the third class 70 – 79 to which the mode $\hat{X}$ belongs and it's given by

$$\hat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times w = 70 + \left(\frac{(12 - 8)}{(12 - 8) + (12 - 10)} \right) \times 10 = \mathbf{76.667} \quad (0.75)$$

4. Compute the variance and standard deviation of the exam scores.

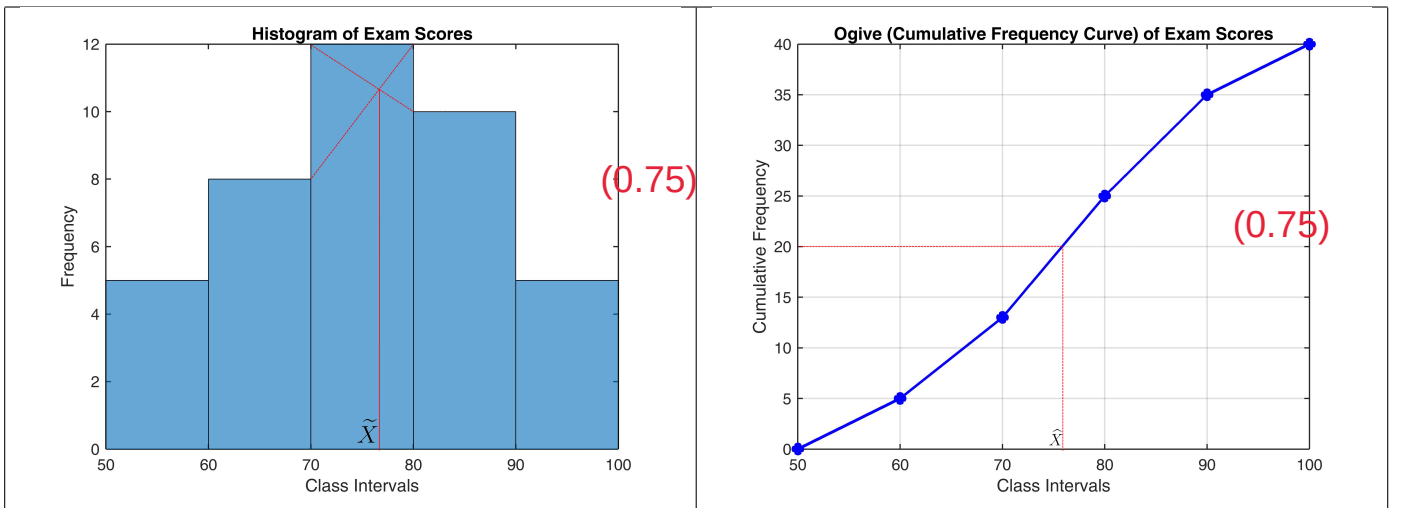
We have $\mu = \bar{X}$ and the variance is given by

$$\sigma^2 = Var(X) = \frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2 = \frac{5790}{40} = \mathbf{144.75} \quad (0.75)$$

therefore, the standard deviation of the exam scores is

$$\sigma = \sqrt{\frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2} = \sqrt{144.75} = \mathbf{12.0312}. \quad (0.25)$$

5. Construct a histogram to visualize the frequency distribution of the exam scores and the Ogive plot of the above distribution then determine the mode and median of the series graphically.



6. Determine the 10th, 25th, 50th, 75th, and 90th percentiles of the exam scores.

The 10th percentiles is also the 1st decile, it's located in the class 50 – 59 and given by

$$P_{10} = D_1 = L_1 + \left(\frac{\frac{n}{10} - Cf_{d_1}}{f_{d_1}} \right) \times w = 50 + \left(\frac{\frac{40}{10} - 0}{5} \right) \times 10 = \mathbf{58}$$

and the 25th, 50th and 75th percentiles are also the 1st quartile, the median and the third quartile respectively , they are respectively located in the classes 60 – 69, 70 – 79 and 80 – 89. They're given by

$$P_{25} = Q_1 = L_1 + \left(\frac{\frac{n}{4} - Cf_{q_1}}{f_{q_1}} \right) \times w = 60 + \left(\frac{\frac{40}{4} - 5}{8} \right) \times 10 = \mathbf{66.25}.$$

$$P_{50} = Q_2 = \tilde{X} = L_1 + \left(\frac{\frac{n}{2} - Cf_{q_2}}{f_{q_2}} \right) \times w = 70 + \left(\frac{\frac{40}{2} - 13}{12} \right) \times 10 = \mathbf{75.833}.$$

$$P_{75} = Q_3 = L_1 + \left(\frac{\frac{3n}{4} - Cf_{q_3}}{f_{q_3}} \right) \times w = 80 + \left(\frac{\frac{3 \times 40}{4} - 25}{10} \right) \times 10 = \mathbf{85}.$$

The 90th percentiles belongs to the class 90 – 99 and given by

(0.5x5)

$$P_{90} = D_9 = L_1 + \left(\frac{\frac{9n}{10} - Cf_{d_9}}{f_{d_9}} \right) \times w = 90 + \left(\frac{\frac{9 \times 40}{10} - 35}{5} \right) \times 10 = \mathbf{92}$$

7. Determine whether the distribution of exam scores is positively skewed, negatively skewed, or symmetrical.

By computing pearson's coefficient of skewness

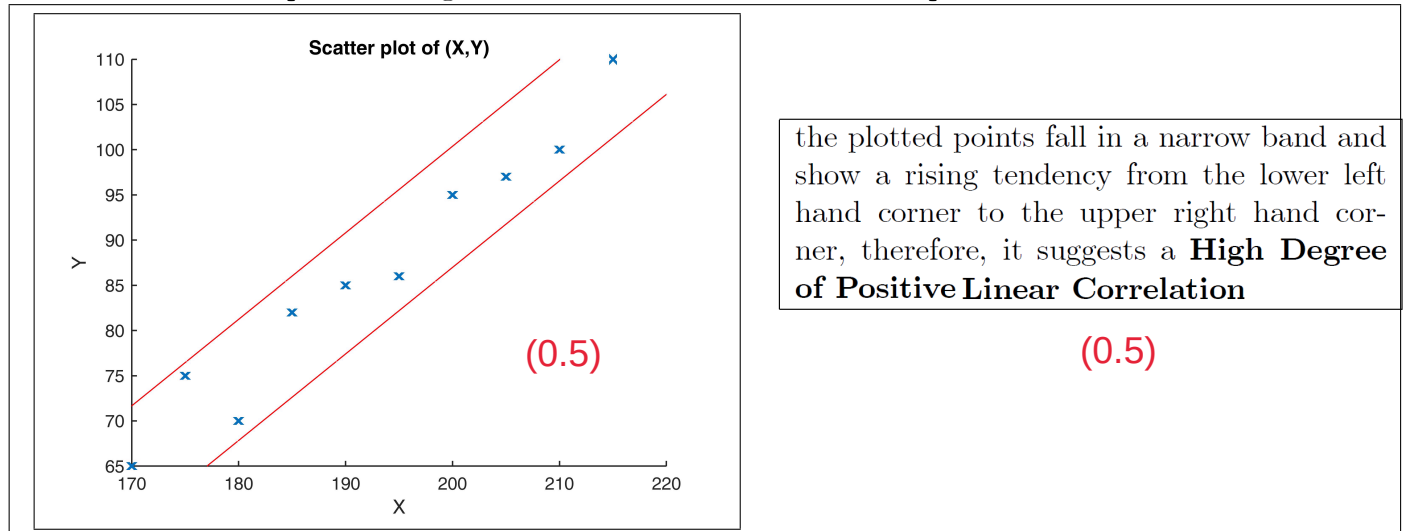
$$sk = \frac{3(\bar{X} - \tilde{X})}{s} = \frac{3 \times (75 - 75.833)}{12.0312} = \mathbf{-0.20771} \quad (0.5)$$

we can observe that its value is close to 0 which let us conclude that the distribution of the exam scores is almost symmetrical (0.25)

8. Consider the following results:

X	170	175	180	185	190	195	200	205	210	215
Y	65	75	70	82	85	86	95	97	100	110

- a. Draw the scatter plot of the given data. Describe the relationship between X and Y.



b. Calculate the correlation coefficient and interpret the obtained result.

The above scatter plot suggested the presence of linear correlation, hence, we choose pearson's coefficient of correlation to measure it.

We have

$$\sum x_i = 1925, \sum y_i = 865, \sum (x_i - \bar{X})^2 = 2062.5, \sum (y_i - \bar{Y})^2 = 1806.5, \sum (x_i - \bar{X})(y_i - \bar{Y}) = 1887.5.$$

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y} = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum (x_i - \bar{X})^2} \sqrt{\sum (y_i - \bar{Y})^2}} = \frac{1887.5}{\sqrt{2062.5} \times \sqrt{1806.5}} = \mathbf{0.97785} \quad (1.5)$$

which indicates the existence of a strong positive linear correlation between the two variables. (0.25)

Exercise 3. 1- Set

$$Z = X - \frac{Cov(X, Y)}{\sigma_Y^2} Y$$

where X and Y are two correlated variables, σ_X^2 and σ_Y^2 are their variances.

Using Z , show that

$$|Cov(X, Y)| \leq \sigma_X \sigma_Y$$

By definition, we have

$$\begin{aligned} 0 &\leq Var(Z) = Cov(Z, Z) = Cov\left(X - \frac{Cov(X, Y)}{\sigma_Y^2} Y, X - \frac{Cov(X, Y)}{\sigma_Y^2} Y\right) \\ &= \underbrace{Cov(X, X)}_{\sigma_X^2} - 2 \frac{Cov(X, Y)}{\sigma_Y^2} Cov(X, Y) + \frac{Cov(X, Y)^2}{(\sigma_Y^2)^2} \underbrace{Cov(Y, Y)}_{\sigma_Y^2} \\ &= \sigma_X^2 - 2 \frac{Cov(X, Y)^2}{\sigma_Y^2} + \frac{Cov(X, Y)^2}{(\sigma_Y^2)^2} \sigma_Y^2 = \sigma_X^2 - \frac{Cov(X, Y)^2}{\sigma_Y^2} \geq 0 \\ &\implies \frac{Cov(X, Y)^2}{\sigma_Y^2} \leq \sigma_X^2 \implies Cov(X, Y)^2 \leq \sigma_X^2 \sigma_Y^2 \\ &\implies |Cov(X, Y)| \leq \sigma_X \sigma_Y. \quad Q.E.D. \end{aligned} \quad (1.5)$$

2- Let's consider a set of observations $x_1, x_2, \dots, x_n$. Define a function $S(m)$ that represents the sum of absolute deviations from a value m

$$S(m) = \sum_{i=1}^n |x_i - m|$$

Show that $S(m)$ is minimized when m is the median of the dataset.

without loss of generality, suppose that $x_1 < x_2 < \dots < x_n$. If $m < x_1$, then

$$S(m) = \sum_{i=1}^n |x_i - m| = \sum_{i=1}^n (x_i - m). \quad (1)$$

As m increases, each term of (1) decreases until m reaches x_1 , therefore $S(x_1) < S(m)$ for all $m < x_1$.

Now suppose that $x_k \leq m \leq m + d \leq x_{k+1}$. Then

$$\begin{aligned} S(m + d) &= \sum_{i=1}^k (m + d - x_i) + \sum_{i=k+1}^n (x_i - m - d) = dk + \sum_{i=1}^k (m - x_i) - (n - k)d + \sum_{i=k+1}^n (x_i - m) \\ &= d(2k - n) + \sum_{i=1}^n |x_i - m| = d(2k - n) + S(m) \end{aligned}$$

so $S(m+d) - S(m) = d(2k - n)$. This is negative if $k < \frac{n}{2}$, zero if $k = \frac{n}{2}$, and positive if $k > \frac{n}{2}$. Thus, on the interval $[x_k, x_{k+1}]$

$$S(m) \text{ is } \begin{cases} \text{decreasing} & \text{if } k < \frac{n}{2} \\ \text{constant} & \text{if } k = \frac{n}{2} \\ \text{increasing} & \text{if } k > \frac{n}{2} \end{cases} \quad (1.5)$$

From here, it shouldn't be too hard to show that $S(m)$ is minimal when m is the median of the dataset $\{x_1, x_2, \dots, x_n\}$.